

IMMS

SMART TRACKING,
SAFE LIVING

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Collecting Requirements

- **Technique 1: Questionnaire**

We made a questionnaire using Google Forms and gave them to users to understand what they want and need from the system. We shared the form with students in our university through WhatsApp groups, and we received 16 responses. For example^[Q], we asked "What features should IMMS have?" many users said they want to know the blood pressure and track chip location. We also asked, "Do you think this system is important in our daily life?". These questions helped us to know the most needed features and how the system should work.

- **Technique 2: Research**

We searched online and asked AI models. We studied how they work and what features they have. For example, we saw that some systems store personal and health data for a long period of time, and they notify user in critical situations via SMS or email. This gave us ideas to add these features to our system. We also read articles to learn what users need in a health system. This helped us understand both functional and non-functional requirements.

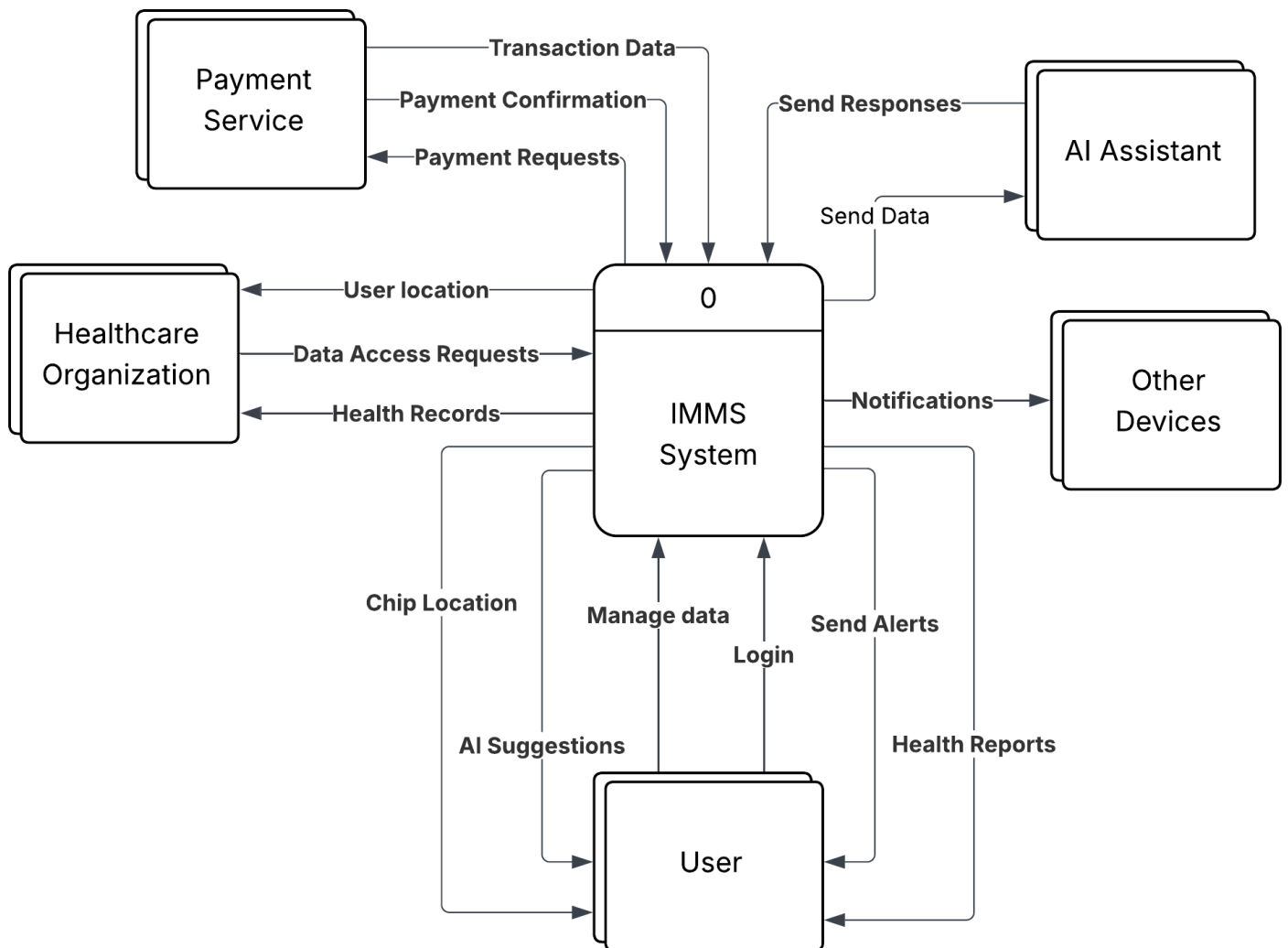
Functional Requirements

- Store personal and health data in a database.
- provide real-time communication between the microchip and external services (e.g., payment, identification).
- Track vital signs (Blood pressure, Pulse rate, Body temperature and Blood oxygen).
- Send alerts to user when something is abnormal.
- Provide an AI assistant.
- Provide a help menu that contains instructions on how to use the system.
- Track the sleeping time.
- Calculate the number of steps and burned calories.
- Send reminder when it is the time to take a pill.
- Connect up-to 3 microchips to a single account.
- Provide real-time location.

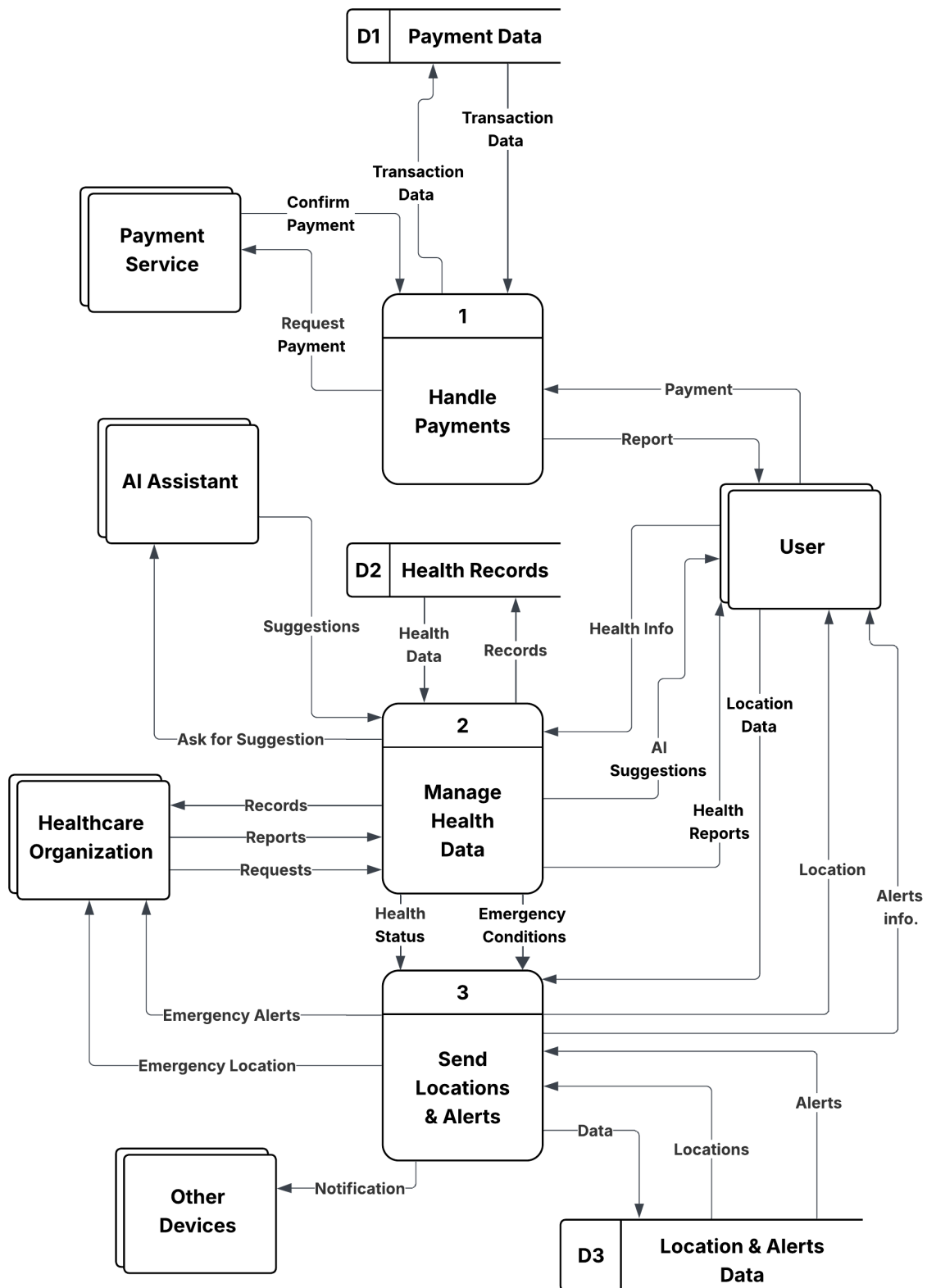
Non-Functional Requirements

- The system should respond to user actions within 1 second.
- The system should be available 24/7.
- All data must be encrypted.
- Only authorized users should be able to access and modify data.
- Supporting multiple platforms (mobiles, desktop devices, etc).
- Real-time data from microchips should be updated quickly without delay.
- The system must be scalable to handle many users (up to 1000 concurrent users).
- Ensure high availability and reliability.
- The system should follow the security standards.
- The system should Follow the legal and ethical standards.
- The system should be under regular maintenance and updates.

Data Flow Diagram (Context Diagram / Level 0)



Data Flow Diagram (Level 1)



Structured English (for process 3)

```
IF location data received THEN  
    send location data to D3  
    IF bad health status THEN  
        send location to user  
        send alerts to user  
        IF emergency condition THEN  
            send emergency location  
            send emergency alerts  
        ENDIF  
    ELSE  
        send location to user  
    ENDIF  
ELSE  
    no action required  
ENDIF
```

Note: Sending emergency alerts or location means sending the data to a healthcare organization.

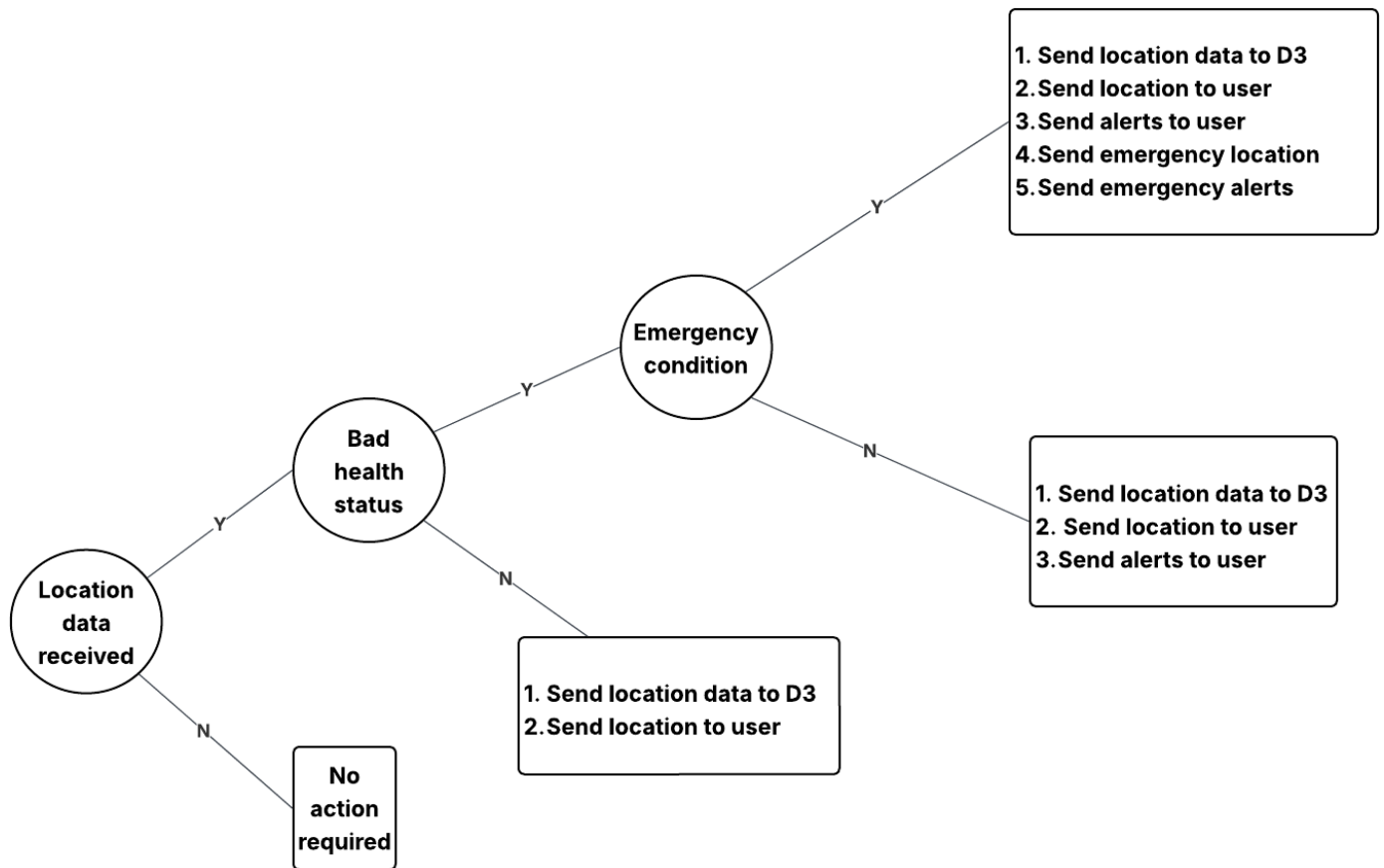
Decision Table (for process 3)

Condition	R1	R2	R3	R4	R5	R6	R7	R8
Location data received	Y	Y	Y	Y	N	N	N	N
Bad health status	Y	Y	N	N	Y	Y	N	N
Emergency condition	Y	N	Y	N	Y	N	Y	N
Actions								
Send data to D3	X	X	X	X				
Send emergency alerts	X							
Send emergency location	X							
Send alerts to user	X	X						
Send location to user	X	X	X	X				
No action required					X	X	X	X

Decision Table (Condensed Version)

Condition	R1	R2	R3	R4
Location data received	Y	Y	Y	N
Bad health status	Y	Y	N	-
Emergency condition	Y	N	-	-
Actions				
Send data to D3	X	X	X	
Send emergency alerts	X			
Send emergency location	X			
Send alerts to user	X	X		
Send location to user	X	X	X	
No action required				X

Decision Tree (for process 3)



Appendix

- **Appendix A: Google Forms Questions**

We distributed the following questionnaire using Google Forms to students at our university via WhatsApp groups. We received a total of 16 responses.

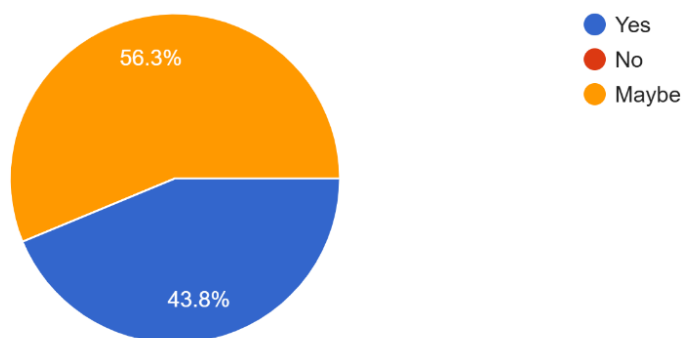
******For full questionnaire: <https://forms.gle/q8fTF19ypeaCdroB7>

Questionnaire Responses

After collecting responses. The following are statistics about important questions that helped us to collect the functional and non-functional requirements.

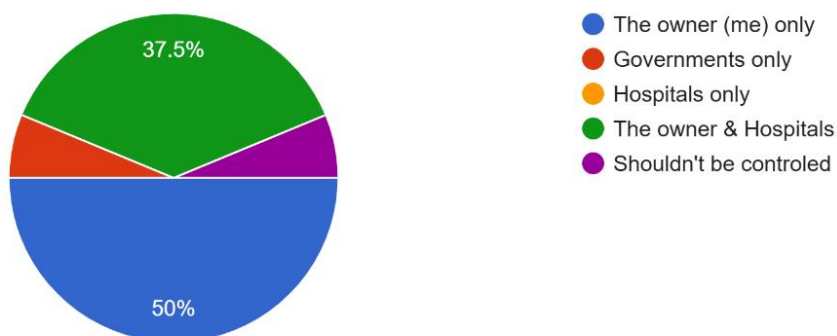
Can implanted microchips replace ID cards and passports?

16 responses



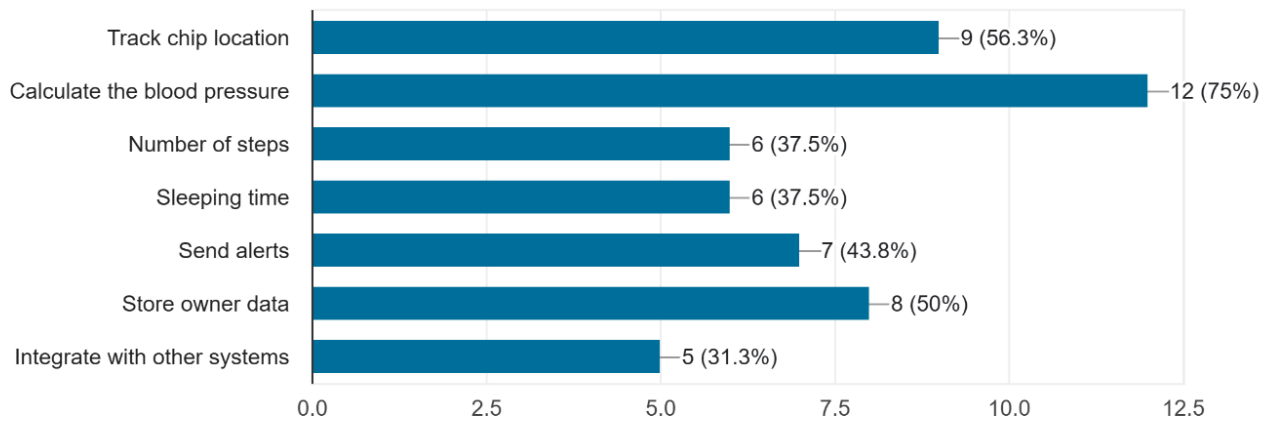
Who should be able to control the microchip data?

16 responses



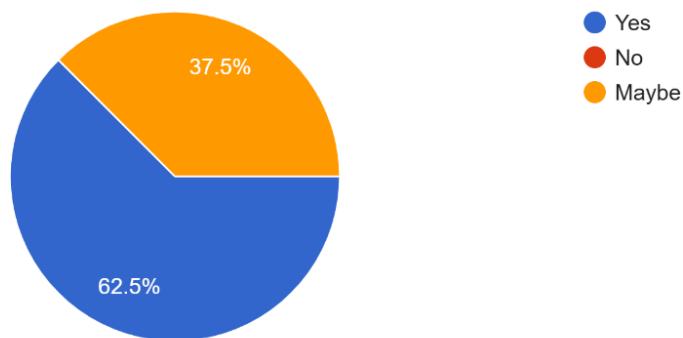
What features do you think IMMS should have?

16 responses



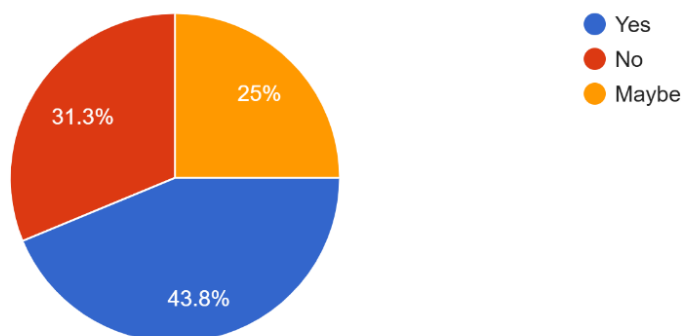
Do you think the system should have an AI assistant?

16 responses



Should the system be able to connect multiple microchips to a single account?

16 responses



- **Appendix B: Users Opinions**

In the form we also included 2 open questions, where users can feel free to share their opinion.

Q1: What advantages do you see about this system?

1. Smart technology powered by AI in our hands would be a good assistant in daily life, being updated of body changes.
2. storing critical health data for emergencies, speeding up response time during accidents or unconscious states.
3. Real time data update / management.
4. Easier identification, keeping track of medical records, facilitates secure transactions.
5. Help user to track his/her health activities.
6. It's helpful.
7. Accurate and Fast Identification.

Q2: What disadvantages do you see about this system?

1. Exploitation and misuse, security risks, threat to our health (cancer risk).
2. Anonymous object installed under the skin, Easy target for cyber-attacks.
3. data can be misused if not properly protected, cyberattacks could compromise sensitive personal or medical information.
4. Safety issues.
5. Cybersecurity Risks.
6. If it's not secure enough I would be afraid to use it
7. Easy to track people with.

- **Appendix C: Research**

To gather more information about this system we used two AI models, ChatGPT and Gemini. We also used the google search engine to find extra resources and examples related to our system.